

Text-, Image-, and Multimodal Fake News Classification: Baselines and Findings

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1 Introduction

This project explores multimodal fake news detection using the *Fakeddit* dataset, which provides three label granularities. An initial missing-data audit shows that `linked_submission_id` has the largest fraction of missing values ($\sim 70\%$), while `domain`, `num_comments`, and `upvote_ratio` each have $\sim 30\%$ missing; most other columns are nearly complete. These observations inform our pre-processing.

We study three settings: **text-only**, **image-only**, and **multimodal** (text+image) classification. Due to computational limits, all models are trained on at most 15k training examples, with **validation/test** at 1.5k each, while preserving the class shares.

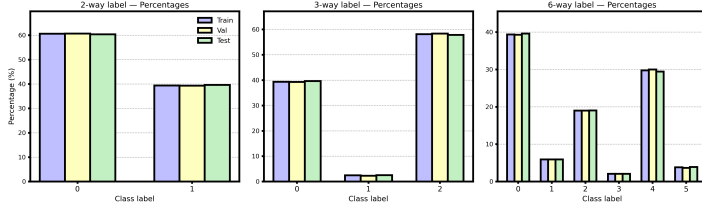


Figure 1: Label distributions across train/validation/test.

Figure 1 summarizes the class distributions in the original dataset. The 3-way and 6-way spaces remain highly imbalanced, which motivates macro-F1 as the primary metric. The shares match our Part 1 EDA. For all experiments and sampling, we maintain these distributions per split (train/validation/test) via stratified procedures to ensure comparable evaluation.

2 Text-Only Baseline

2.1 Overview

We use three splits that mirror our earlier EDA, but cap training at 15,000 samples (validation/test 1,500 each) to keep all baselines comparable under limited compute. Each row includes a normalized title (`clean_title`); a timestamp (`created_utc`); engagement fields (`num_comments`, `score`, `upvote_ratio`); and the label columns `2_way_label`, `3_way_label`, `6_way_label`. Raw data also contains source-identifying fields such as `author`, `domain`, `subreddit`, and identifiers/URLs.

2.2 Leakage Investigation

We first trained a diagnostic variant that *included* source-identifying fields (e.g., domains/subreddits). It yielded near-100% scores, confirming the model could infer labels from sources rather than content. Based on this evidence, we excluded these fields from the final pipeline: `author`, `domain`, `subreddit`, `id`, `image_url`, `linked_submission_id`. All reported results are therefore leak-safe and reflect content understanding.

2.3 Data, Pre-processing, and Model

We use `clean_title` and derive compact numeric cues (length, punctuation counts, capitalization ratio, posting hour/weekday, engagement). For text, we concatenate TF-IDF word n -grams (1–2) and character n -grams (3–5), then append scaled numeric features. We train a linear large-margin classifier using `SGDClassifier` with hinge loss (linear SVM), class balancing, and early stopping. We sweep

$$\alpha \in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}\},$$

and select $\alpha = 10^{-4}$ for all tasks. Final models are retrained on train+val and evaluated on test.

Table 1: Validation (chosen α) and test performance for the leak-safe text+numeric baseline.

Label	Validation		Test (Macro)			
	F1	Acc.	Prec.	Rec.	F1	Acc.
2-way	0.8208	0.8273	0.7950	0.8071	0.7862	0.7873
3-way	0.6812	0.7820	0.8164	0.7010	0.7314	0.7820
6-way	0.5690	0.7893	0.5990	0.6176	0.5831	0.7347

We trained two baselines: (i) *text+numeric* (TF-IDF plus `num_comments`, `upvote_ratio`, `score`, time from `created_utc`, `hasImage`), which gave the best macro-F1 in Table 1; and (ii) a *pure text* model (word/char n -grams + light text stats) to match common practice and our multimodal setup.

For 2-way, top weights are purely lexical. For 3-way, `num_comments`/`hasImage` appear among top features. For 6-way, numeric cues (`num_comments`, `upvote_ratio`, `hasImage`) matter more, explaining the larger gap. This depicts that engagement features help most for fine-grained labels, but a strong, leak-safe text-only baseline remains competitive and is our primary report.

2.4 Results

Figures 2 and 3 report the results on the text-only model. Compared with the text+numeric baseline, the text-only model is uniformly weaker on test.

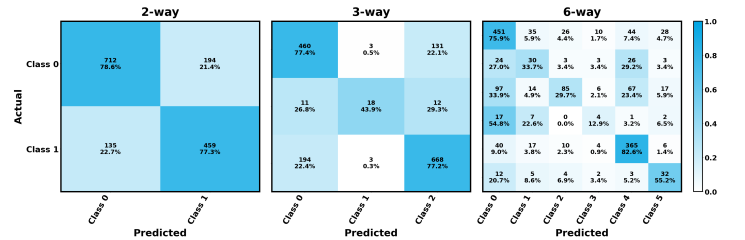


Figure 2: Normalized confusion matrices for 2-way, 3-way, and 6-way (test), text.

2.5 Discussion

On test, macro-F1 declines from 0.7862 $\rightarrow$ 0.7742 (2-way), 0.7314 $\rightarrow$ 0.6939 (3-way), and 0.5831 $\rightarrow$ 0.4645 (6-way), with parallel accuracy drops. On validation, text-only macro-F1 is

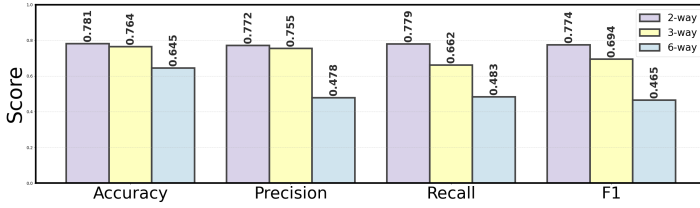


Figure 3: Text test metrics across tasks (accuracy, macro-precision, macro-recall, macro-F1).

0.7997/0.6832/0.4650 (2/3/6-way) versus 0.8208/0.6812/0.5690 for text+numeric—nearly tied on 3-way but clearly weaker on 6-way. Confusion matrices show minority classes collapsing into major ones (e.g., 3-way class 1; several 6-way rarities), while 6-way class 4 remains well captured. Overall, leak-safe *text* is strong for 2-way and competitive for 3-way, but lightweight engagement features provide decisive gains for the fine-grained 6-way task. The strongest signals come from TF-IDF n -grams; numeric features add modest complementary value once source-identifying fields are excluded. The best results we saw were from training the text-only baseline using the text+numeric model on the complete dataset, which led to above 80 percent F1 and accuracy, but for fair comparison we consider the same model trained on a smaller train set of 15k in the report.

3 Image-Only Baseline

3.1 Overview

We train a compact image-only classifier that uses the post image plus two auxiliary numeric features (original width, height). To avoid leakage, we exclude all source-identifying fields (IDs, domains, subreddits, URLs). The pipeline: download/reuse images, filter unusable files, standardize spatial size and channel stats, then train a lightweight CNN with a small numeric MLP head. Training is capped at 15k images (val/test 1.5k each) while preserving label proportions.

3.2 Data, Pre-processing, and Model

Pre-processing. RGB conversion, resize+center-crop to 224×224, normalization with train-split channel statistics. The original (width, height) are standardized and passed as two numeric features.

Model. Three Conv-ReLU blocks with stride 2 + global average pooling ($\mathbb{R}^{128}$). Numeric MLP ($2 \rightarrow 32 \rightarrow 16$). Concatenate ($128+16$) then a $144 \rightarrow 64 \rightarrow C$ head where $C \in \{1, 3, 6\}$. BCEWithLogitsLoss for 2-way; CrossEntropyLoss otherwise.

Training. Adam (10^{-3}), batch size 32, 10 epochs.

3.3 Results

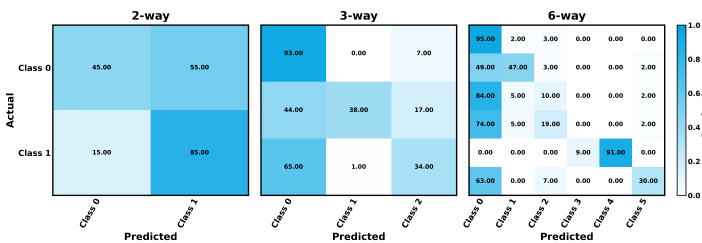


Figure 4: Normalized confusion matrices for 2-way, 3-way, and 6-way (test), image-only.

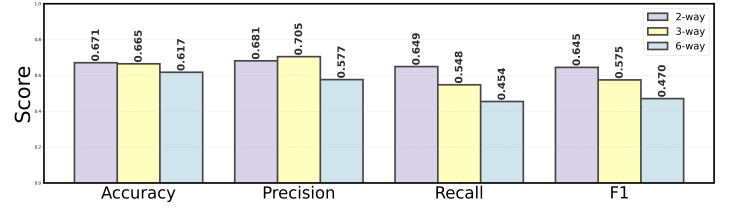


Figure 5: Image-only test metrics across tasks (Accuracy, macro-Precision, macro-Recall, macro-F1).

On the held-out test set, macro-F1 is 0.645 (2-way), 0.575 (3-way), and 0.471 (6-way) (see Fig. 4 and Fig. 5). On validation, we tracked accuracy/loss, which are approximately 0.670/0.617 for 2-way, 0.660/0.684 for 3-way, and 0.611/0.983 for 6-way.

3.4 Discussion

Overall accuracy sits around the high-0.6s for the 2-way task (Fig. 5) with macro-Precision and macro-Recall closely matched, indicating balanced positive/negative decisions from the visual stream alone. For 3-way and especially 6-way, macro-Recall drops more than precision, which suggests the model is conservative on minority classes—consistent with the confusion matrices where rare classes bleed into dominant ones (Fig. 4). In practice, this means the image encoder captures broad cues of credibility but struggles to separate fine-grained categories that are visually similar (e.g., satire vs. true vs. miscaption). Strengthening the encoder (CLIP/ViT), adding targeted augmentations, or using class-balanced losses is likely to lift recall without sacrificing precision.

4 Multimodal

4.1 Overview

We implemented a multimodal baseline inspired by the Fakeddit work [1]. The model fuses textual and visual signals: a GRU-based text encoder and a compact CNN-based image encoder. Training uses 15k (image,text) pairs with stratified sampling (val/test 1.5k/1.5k), preserving the class distribution reported in our EDA.

4.2 Data, Pre-processing, and Model

Text tower. Cleaned titles are tokenized, embedded (pretrained vectors), and encoded with a bidirectional GRU (hidden dim 128); sequence outputs are mean-pooled.

Image tower. Same as image-only: three Conv-ReLU blocks with stride 2, global average pooling to $\mathbb{R}^{128}$.

Fusion. Concatenated text and image vectors pass through a two-layer MLP (256 hidden units, ReLU, dropout 0.2) and a task-specific head ($C \in \{2, 3, 6\}$). Losses: BCEWithLogitsLoss (2-way) / CrossEntropyLoss (3-/6-way). Optimizer: AdamW (10^{-3} , weight decay 10^{-4}). Batch size 32, 10 epochs, AMP enabled.

4.3 Results

On the held-out test set, the multimodal model attains macro-F1/accuracy of 0.704/0.705 (2-way), 0.677/0.689 (3-way), and 0.402/0.490 (6-way) (see Fig. 6 and Fig. 7). The best validation checkpoints with early stopping reached:

F1 $\approx$ 0.707 at val. loss $\approx$ 0.634 (2-way)

F1 $\approx$ 0.643 at val. loss $\approx$ 0.674 (3-way)

F1 $\approx$ 0.401 at val. loss $\approx$ 1.424 (6-way)

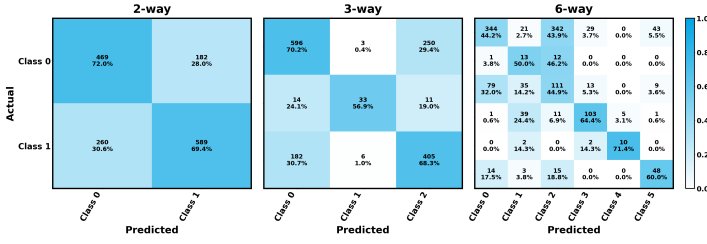


Figure 6: Normalized confusion matrices for 2-way, 3-way, and 6-way (test), multimodal.

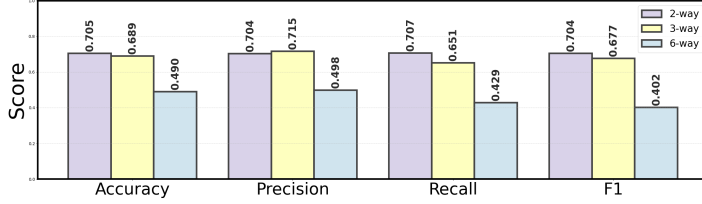


Figure 7: Multimodal test metrics across tasks (Accuracy, macro-Precision, macro-Recall, macro-F1).

4.4 Discussion

Fusion yields the most consistent behavior on the 2-way task: the bar chart indicates accuracy, precision, recall, and F1 all around ≈ 0.70 – 0.71 , showing that text and image cues complement one another without obvious trade-offs. For 3 ways, macro-precision (≈ 0.72) exceeds macro-recall (≈ 0.65), implying that the model is confident when it predicts minority classes but still misses a portion of them, visible in Fig. 6 where off-diagonal mass is concentrated from minority classes into the dominant one. On 6-way, accuracy settles near ≈ 0.49 with macro-F1 ≈ 0.40 : the model recognizes the most frequent categories but confuses under-represented ones. This pattern suggests two levers: (1) stronger encoders (e.g., CLIP/ViT for images and domain-adapted text encoders) to separate visually/semantically similar categories, and (2) imbalance-aware training (class-balanced sampling, reweighting, or focal losses) to lift macro-Recall without eroding precision. Since we restricted training to 15k pairs, we expect measurable gains from scaling data or using pretrained multimodal features.

5 Discussion & Conclusion

5.1 Cross-Modal Comparison

Across tasks, **text** models are strongest, **image** models are weakest, and **multimodal** fusion helps mainly in 2–3 way but not in 6-way. On the held-out test set, the leak-safe *text+numeric* baseline attains macro-F1 ≈ 0.786 (2-way), 0.731 (3-way), and 0.583 (6-way). The *text-only* variant trails at $0.774/0.694/0.465$, confirming that lightweight engagement signals (e.g., comment counts, upvote ratio) add value—especially for the fine-grained labels. The *image-only* CNN lags at $\sim 0.645/0.575/0.471$, while the *multimodal* GRU+CNN reaches $\sim 0.704/0.677/0.402$ (macro-F1 for 2/3/6-way). Validation checkpoints show similar ordering (e.g., multimodal val F1 $\approx 0.707/0.643/0.401$).

5.2 Why These Patterns?

(i) Information content. Short titles carry strong topical/stylistic cues that TF-IDF models exploit well; adding simple engagement features boosts separability for rare 6-way classes (e.g., class frequency/interest patterns), narrowing ambiguity where text alone is thin.

(ii) Visual difficulty. Many 6-way distinctions are subtly visual (e.g., repurposed or misleading context) and hard for a small CNN trained from scratch on 15k examples; consequently the image-only model under-recalls minorities and fusion inherits this weakness when the visual stream is noisy or weakly aligned with the title.

(iii) Class imbalance. The 3- and 6-way spaces are highly imbalanced; macro-F1 exposes that models collapse minority classes into dominant ones (as seen in confusion matrices). Without imbalance-aware losses/sampling, fusion and vision suffer most in recall.

(iv) Capacity & pretraining. Our encoders are intentionally lightweight to represent the baseline methods (linear TF-IDF; small CNN/GRU) and trained on capped data (15k pairs). Stronger pretrained towers (e.g., CLIP/ViT for images; domain-adapted transformers for text) and longer training would likely improve 3-/6-way separation and make fusion consistently superior.

5.3 Takeaways

- **Text first:** A leak-safe text baseline is a strong reference; adding a few engagement features lifts 6-way substantially.
- **Vision helps with scale:** Our small CNN underperforms; expect larger gains with pretrained vision encoders and class-balanced training.
- **Fusion needs stronger parts:** Multimodal is competitive in 2–3 way but trails in 6-way because the vision tower is the bottleneck and minority recall is low.

5.4 Conclusion & Next Steps

Overall, text models set the bar; simple engagement features further improve fine-grained detection; and multimodal fusion becomes reliably better once each modality is strong on its own. Immediate next steps are: (1) replace the CNN/GRU with pretrained CLIP/ViT and a transformer text encoder; (2) use imbalance-aware objectives (class-balanced or focal loss, reweighting, and calibrated thresholds) to raise macro-recall; (3) scale training pairs beyond 15k and apply targeted augmentations; and (4) explore cross-modal consistency losses to penalize mismatched image–text pairs.

6 Assumptions & Result Reproduction

For running the text-only and image-only notebooks, the dataset paths must be updated within the respective notebooks. The original dataset and TSV files are available on Google Drive, and detailed steps for downloading them are provided in the notebooks. The same images downloaded in the image-only notebook are reused in subsequent experiments.

For the existing multimodal setup, it is assumed that the 15k images have already been downloaded from the previous (image-only) step. Only the file paths to these images need to be updated within the script before execution.

References

- [1] B. D. Horne, J. Nørregaard, and S. Adali. *Fakeddit: A New Multimodal Benchmark Dataset for Fake News Detection*. arXiv:2112.04831, 2021. <https://arxiv.org/abs/2112.04831>